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CHAPTER 1 • NUMBER • TOPIC 2

Sets & Venn Diagrams

Learn four symbols, then every question is listing, counting or filling regions — always from the middle out.



$$\frac{3}{10} = 0.3 =$$

30%

6

VARIANTS

25

PAST-PAPER PARTS

32

PRACTICE QUESTIONS

YOU WILL MASTER

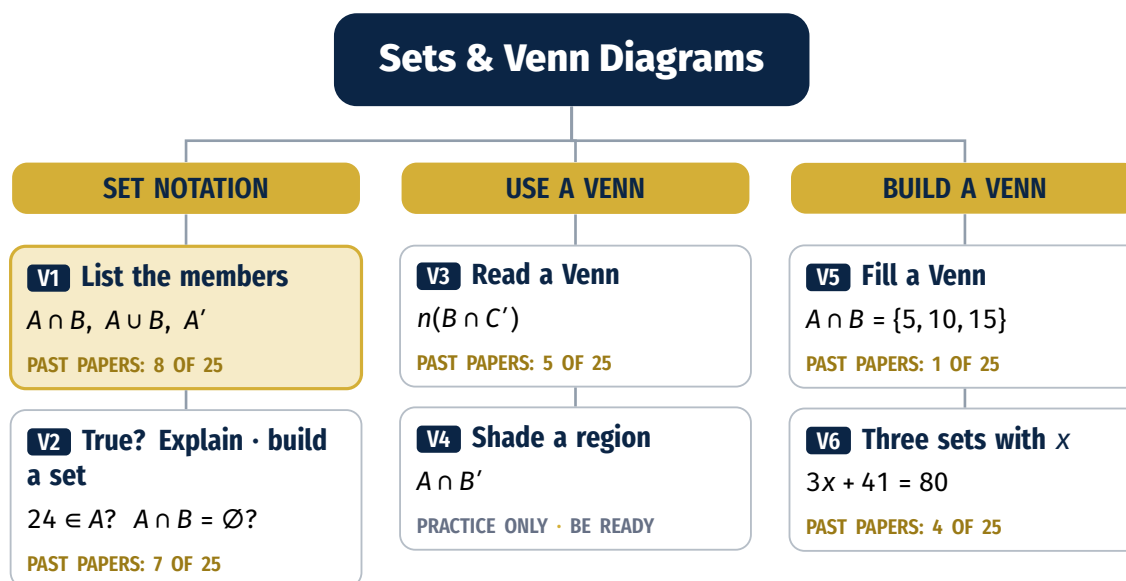
V1 List the members V2 True? Explain V3 Read a Venn

V4 Shade a region V5 Fill a Venn V6 Three sets with x

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Sets & Venn Diagrams at a glance



Your notes
**DR ESLAM'S RULE****FILL FROM THE MIDDLE OUT**

Centre first, then the overlaps, then the “only” parts, then outside.

How often does each variant come up?

18 past-paper questions (Jan 2020 → Jun 2025) hold 25 set parts; 32 practice parts — grouped by type.

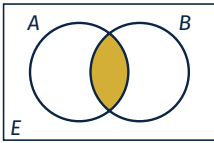
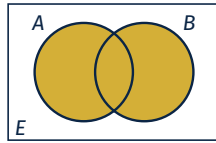
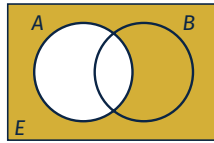
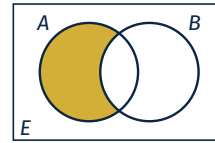
	PAST PAPERS (OUT OF 25)	PRACTICE PARTS
V1 List the members	8	10
V2 True? Explain • build a set	7	7
V3 Read a Venn	5	5
V4 Shade a region	0	2
V5 Fill a Venn	1	5
V6 Three sets with x	4	3

WHAT THIS TELLS YOU A set question is on most Paper 1s: listing $n, \cup, '$ (easy marks) plus a “reason” part. The 6-mark three-circle x question appears every year or so.

BEFORE YOU START

The symbols



Your notes
 $A \cap B$ "and" $A \cup B$ "or" A' "not A" $A \cap B'$ "A only"

E
universal set
Everything in the
question —
nothing outside
it.

$\in / \notin$
 $9 \in A, 24 \notin E$
"is a member of"
/ "is not".

$\emptyset$
 $A \cap C = \emptyset$
The empty set —
no members.

$n(A)$
 $n(A) = 6$
The **number** of
members.



Words to symbols

both / and

$\cap$
in A **and** in B

either / or

$\cup$
in A, in B, or in both

not / neither

$'$
everything outside



CALCULATOR FIRST, THEN CHECK

Write E out in full first, then each set in full: "multiples of 3" in $\{9, \dots, 20\}$ is $\{9, 12, 15, 18\}$.
Tick each member as you place it — every member of E goes in exactly one region.

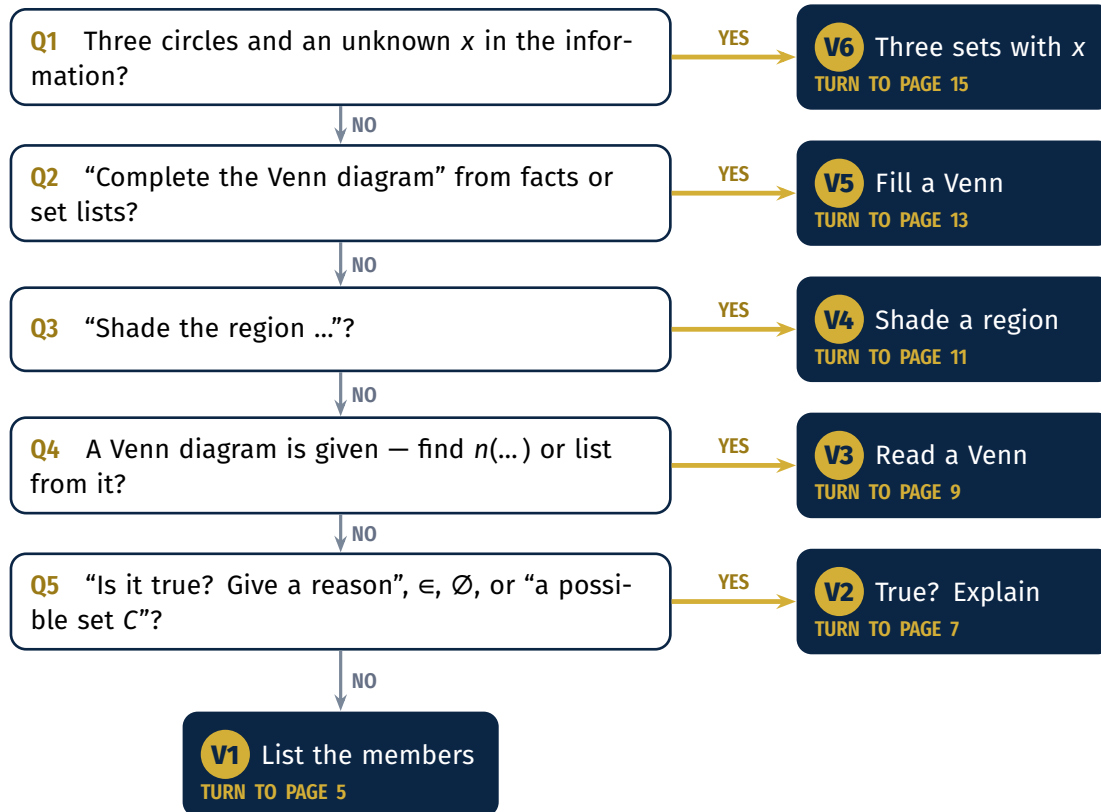
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

List the Members

Write every set in full, then pick out $\cap$, $\cup$ or $'$.

SEEN IN PAST PAPERS

8 / 25

PRACTICE QUESTIONS: 10



Your notes

Came up in: Jan 2020 P1HR Q4 • Jan 2021 P1H Q4 • Jan 2021 P2HR Q3 • May 2021 P2H Q4 • Jun 2022 P2HR Q4 • May 2024 P1H Q7 • Jun 2024 P2HR Q4 • Jun 2025 P1HR Q5

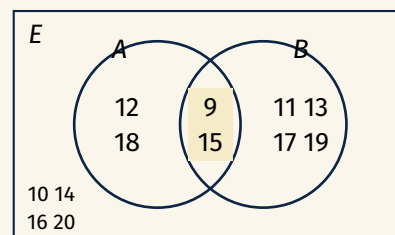
? What is it?

In plain words

Sets are described in words (“odd numbers”). First **write each set in full** — only numbers from E .

Then: $\cap$ = in **both**, $\cup$ = in **either** (no repeats), $'$ = in E but **not** in A .

IT LOOKS LIKE

 $A \cap B$ $A \cup B$ A' $W \cap G'$ 

How do I do it?

1 Write E

List every member of the universal set.

2 Write each set

Only members of E that fit the rule.

3 Pick out

 $\cap$ both • $\cup$ either • $'$ not.

4 Curly brackets

Answer as a set: $\{9, 15\}$ — each member once.

WRITE EVERY SET IN FULL FIRST

Then it is just matching.

V1 List the Members

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"List the members of the set $A \cap B$ "*→ **Members in both***" $A \cup B$ "*→ **Members in either — write each once***" A' " or " $W \cap G'$ "*→ **Not in A / in W but not in G** **WORKED EXAMPLE****PAST PAPER**

June 2022 • Paper 2HR • Q4(a) • 2 marks

$E = \{9, 10, \dots, 20\}$, $A = \{\text{multiples of } 3\}$, $B = \{\text{odd numbers}\}$. List the members of (i) $A \cap B$ (ii) $A \cup B$.

STEP 2

$A = \{9, 12, 15, 18\}$

◀ multiples of 3 in E **STEP 2**

$B = \{9, 11, 13, 15, 17, 19\}$

◀ odd numbers in E **STEP 3**

$A \cap B = \{9, 15\}$

◀ in both

STEP 3

$A \cup B = \{9, 11, 12, 13, 15, 17, 18, 19\}$

◀ in either, once each

ANSWER (i) $\{9, 15\}$ (ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$ ✓ 8 members, none repeated

**WATCH OUT**✗ $A \cup B = \{9, 12, 15, 18, 9, 11, 13, 15, \dots\}$

✓ each member once

A set never repeats a member.

**EXAM TIP**

Use only numbers from E : 21 is a multiple of 3 but is not in $E = \{9, \dots, 20\}$.

**Check yourself**

calculator first • answers on page 17

1.1

Jan 2021 P1H

$E = \{20, \dots, 29\}$, A odd, B multiples of 3. List $A \cap B$ and $A \cup B$.

Answer:

1.2

Jun 2025 P1HR

$E = \{1, \dots, 12\}$,
 $A = \{2, 4, 6, 8, 10, 12\}$,
 $B = \{3, 6, 9, 12\}$. List
 $A \cup B$.

Answer:

1.3

Jun 2025 P1HR

With the sets in 1.2, list
 B' .

Answer:

V2

VARIANT 2

True? Explain • Build a Set

Decide with a reason that names a member
— or build a set that fits all the conditions.

SEEN IN PAST PAPERS

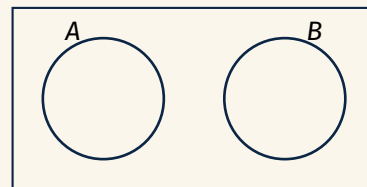
7 / 25

PRACTICE QUESTIONS: 7



Your notes

Came up in: Jan 2020 P1HR Q4(b) • Jan 2021 P2HR Q3(b) • May 2021 P2H Q4(b) • Jun 2022 P2HR Q4(b)(c) • May 2024 P1H Q7(b)(c)
• Jun 2024 P2HR Q4(b)(c) • Jun 2025 P1HR Q5(b)

? What is it?**In plain words****Is it true?** Give the decision**and a member that proves it:** “No — 2 and 3 are in both A and B.”**Build a set C:** meet every condition one at a time; check at the end. $\emptyset$ means **no members**: $A \cap B = \emptyset$ means the circles do not overlap. $A \cap B = \emptyset$: **NO OVERLAP****IT LOOKS LIKE**

$A \cap B = \emptyset?$

$24 \in A?$

$C \cap B' = \{10$

$18\}$

How do I do it?**1 Write the sets**

In full, as in V1.

2 Test the claim

Look for a member that breaks (or proves) it.

3 Reason

Decision + the member: “No, 24 is not in E.”

4 Build a set

Choose members condition by condition; then check every condition.

A REASON NAMES A MEMBER

“Yes/No” alone scores 0.

V2 True? Explain · Build a Set

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Harpreet states that $A \cap B = \emptyset$.
Explain why she is incorrect"

→ **Name a member in both**

"Is it true that $24 \in A$?"

→ **Check E first — is 24 even in it?**

"List the members of one possible set
 C "

→ **Build it condition by condition, then check****WORKED EXAMPLE** **PAST PAPER**

June 2024 · Paper 2HR · Q4(b)(c) · 3 marks

$E = \{1, \dots, 10\}$, $A = \{\text{factors of } 6\}$, $B = \{\text{prime numbers}\}$. (b) Why is $A \cap B = \emptyset$ wrong? (c) C has 4 members; $A \cap C$ and $B \cap C$ have 2 members each and nothing in common. List C .

STEP 1 $A = \{1, 2, 3, 6\}$, $B = \{2, 3, 5, 7\}$ ◀ write the sets

STEP 3 2 and 3 are in both A and B , so $A \cap B \neq \emptyset$ ◀ (b) reason with members

STEP 4 $A \cap C = \{1, 6\}$, $B \cap C = \{5, 7\}$ ◀ (c) avoid 2, 3 (they are in both)

ANSWER (b) 2 and 3 are in both (c) $C = \{1, 5, 6, 7\}$ ✓ each condition checked

WATCH OUT

× "No, because they share numbers."

✓ "No — 2 and 3 are in both A and B ."

Name the members: that is the mark.

EXAM TIP

When you build a set, write the check line: $A \cap C = \{1, 6\}$, $B \cap C = \{5, 7\}$ — 2 members each ✓.

Check yourself

calculator first · answers on page 17

2.1 Jun 2022 P2HR

$E = \{9, \dots, 20\}$,
 $A = \{\text{multiples of } 3\}$. Is it
true that $24 \in A$?

Answer:

2.2 Jan 2020 P1HR

$B = \{b, l, u, e\}$,
 $G = \{g, r, e, y\}$,
 $W = \{w, h, i, t, e\}$. Is
 $B \cap G \cap W = \emptyset$?

Answer:

2.3 Jan 2021 P2HR

$E = \{21, \dots, 30\}$,
 $A = \{22, 24, 26, 28, 30\}$,
 $B = \{21, 24, 27, 30\}$,
 $C = \{23, 25, 29\}$. Write C
in terms of A and B .

Answer:

V3

VARIANT 3

Read a Venn Diagram

Numbers in regions: add the regions that the notation describes.

SEEN IN PAST PAPERS

5 / 25

PRACTICE QUESTIONS: 5



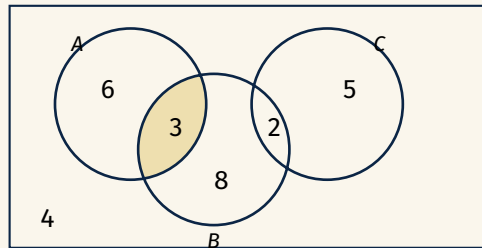
Your notes

Came up in: May 2021 P1H Q19 • Jan 2022 P1HR Q17 • May 2023 P1HR Q2 • Nov 2023 P1H Q1 • Nov 2024 P1H Q4

? What is it?**In plain words**

Each number is **how many** are in that region. $n(A \cup B)$: add **every region inside** A or B.

Shade it in your head (or lightly with pencil), then add the shaded numbers.

IT LOOKS LIKE $n(A \cup B)$ $n(A \cap C)$ $n(B \cap C')$ 

Jan 2022 P1HR Q17: A and C do not overlap.

How do I do it?**1 Read the notation**

$\cup$ either, $\cap$ both, ' outside.

2 Mark the regions

Lightly tick each region that fits.

3 Add

Add the numbers in the ticked regions.

4 Check the total

$n(E)$ is all regions, outside included.

SHADE, THEN ADD

Never add a region twice.

V3 Read a Venn Diagram

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"6, 3, 8 ... represent the numbers of elements"

→ **Add regions — do not list**

"Find $n(B \cap C')$ "

→ **In B, not in C**

" $n(A' \cup B' \cup C')$ "

→ **Everything except the region in all three****WORKED EXAMPLE** **PAST PAPER**

January 2022 • Paper 1HR • Q17 • 4 marks

A only 6, $A \cap B$ 3, B only 8, $B \cap C$ 2, C only 5, outside 4 (A and C do not meet). Find (i) $n(A \cup B)$ (ii) $n(A \cap C)$ (iii) $n(B \cap C')$ (iv) $n(A' \cup B' \cup C')$.

STEP 3 $n(A \cup B) = 6 + 3 + 8 + 2 = 19$

◀ (i) every region in A or B

STEP 3 $n(A \cap C) = 0$

◀ (ii) the circles do not overlap

STEP 3 $n(B \cap C') = 3 + 8 = 11$

◀ (iii) in B, not in C

STEP 4 $n(A' \cup B' \cup C') = 28$

◀ (iv) no region is in all three: everything

ANSWER 19, 0, 11, 28✓ total $6 + 3 + 8 + 2 + 5 + 4 = 28$ **WATCH OUT**✗ $n(A \cup B) = 6 + 3 + 8 = 17$ ✓ $6 + 3 + 8 + 2 = 19$

The 2 is in B (and C) — it belongs to $A \cup B$ too.

EXAM TIP

Find $n(E)$ first — many "" answers are $n(E)$ minus something.

Check yourself

calculator first • answers on page 17

3.1 Dr Eslam

A only 7, $A \cap B$ 4, B only 9, outside 5. Find $n(A \cup B)$.

Answer:

3.2 Dr Eslam

Same diagram: find $n(B')$.

Answer:

3.3 Dr Eslam

Same diagram: find $n(A \cap B')$.

Answer:

V4

VARIANT 4

Shade a Region

Turn the notation into a region: $\cap$ is the overlap, $'$ is outside.

SEEN IN PAST PAPERS

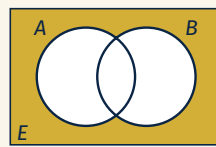
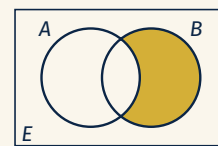
0 / 25

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2


Your notes
? What is it?**In plain words**

Build the region in steps: shade the first part lightly, then keep only what the **next symbol** allows.

$A \cap B'$: inside A , outside B — the “**A only**” crescent.

IT LOOKS LIKE $A \cap B'$ $(A \cup B)'$ $A' \cap B$  $(A \cup B)'$  $A' \cap B$ **How do I do it?****1 Brackets first**

Work out the bracket region, e.g. $A \cup B$.

2 Apply $'$

Swap shaded and unshaded (everything else in E).

3 Apply $\cap$

Keep only regions shaded in **both** parts.

4 Check

Pick a point in your shading: does it fit the words?

$\cap$ = KEEP THE OVERLAP • $'$ = SWAP Build it one symbol at a time.

V4 Shade a Region

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

- “Shade the region $A \cap B'$ ” → **A only**
- “ $(A \cup B)'$ ” → **Outside both circles**
- “Describe the shaded region in set notation” → **Read it: in? out? → $\cap, '$**

WORKED EXAMPLE DR ESAM

Shading · 2 marks

On a two-set Venn diagram, shade the region $A' \cap B$.

- STEP 1** no brackets ◀ brackets first
- STEP 2** A' : everything outside circle A ◀ apply $'$
- STEP 3** keep only the part also in B ◀ apply $\cap$
- STEP 4** a point in “B only” is in B and not in A ✓ ◀ check

ANSWER **B only (the crescent of B outside A)** ✓ see the picture on page A

WATCH OUT

- ✗ shading all of B
- ✓ only the part of B outside A
- A' removes the overlap.

EXAM TIP

Shade clearly and only the region asked
— extra shading loses the mark.

Check yourself

calculator first · answers on page 17

4.1 Dr EslamWhich region is $A \cap B'$?

Answer:

4.2 Dr EslamWhich region is $(A \cup B)'$?

Answer:

4.3 Dr EslamWhich region is $(A' \cup B')'$?

Answer:

V5

VARIANT 5

Fill a Venn Diagram

Fill from the middle out: the overlap first, then the “only” parts, then outside.

SEEN IN PAST PAPERS

1 / 25

PRACTICE QUESTIONS: 5



Your notes

Came up in: Jan 2022 P1H Q7

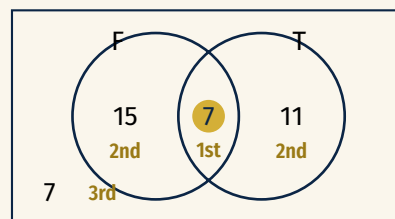
? What is it?**In plain words**

The overlap first — it belongs to both sets. Then each “only” part is the set **minus the overlap**. Then outside = total minus everything inside.

With numbers: 22 football, 7 both → football only $22 - 7 = 15$.

IT LOOKS LIKE

22 football, 18 tennis, 7 both

**How do I do it?****1 Overlap**

Put the “both” members (or number) in the middle.

2 Only parts

Each set minus what is already in the overlap.

3 Outside

What is left of E (or total minus inside).

4 Check

Every member placed once / regions add to the total.

FILL FROM THE MIDDLE OUT Dr Eslam's order.

V5 Fill a Venn Diagram

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Complete the Venn diagram for this information”

→ **Overlap, then only parts, then outside**

A' and B' given

→ **A only = B' minus A' (in A , not in B)**

WORKED EXAMPLE **PAST PAPER**

January 2022 · Paper 1H · Q7 · 3 marks

$E = \{4, \dots, 15\}$, $A \cap B = \{5, 10, 15\}$, $B' = \{7, 8, 9, 11, 12, 13, 14\}$, $A' = \{4, 6, 7, 8, 14\}$. Complete the Venn diagram.

STEP 1 middle: 5, 10, 15

◀ the overlap first

STEP 2 A only = $\{9, 11, 12, 13\}$, B only = $\{4, 6\}$

◀ A only: in B' not A' ; B only: in A' not B'

STEP 3 outside = $A' \cap B' = \{7, 8, 14\}$

◀ in neither

ANSWER A only 9,11,12,13 · both 5,10,15 · B only 4,6 · outside 7,8,14
members placed

✓ all 12

WATCH OUT

✗ writing 22 in the football-only region

✓ $22 - 7 = 15$

22 is the whole circle — the overlap is part of it.

EXAM TIP

Count at the end: the numbers in all regions must add up to $n(E)$.

Check yourself

calculator first · answers on page 17

5.1

Dr Eslam

40 students: 22 play football, 18 tennis, 7 both. How many play neither?

Answer:

5.2

Practice · Hard

25 people each have a cat or a dog or both; 17 have a cat, 15 a dog. How many have both? Only one?

Answer:

5.3

Practice · Easy

50 people: 30 have a cat, 25 a dog, 11 both. How many have neither?

Answer:

V6

VARIANT 6

Three Sets with x

Three circles, one unknown: fill from the centre, write the total as an equation in x .

SEEN IN PAST PAPERS

4 / 25

PRACTICE QUESTIONS: 3



Your notes

Came up in: Jan 2020 P2HR Q17 • Jan 2021 P2H Q16 • May 2023 P1H Q16 • Nov 2024 P2H Q21

? What is it?**In plain words**

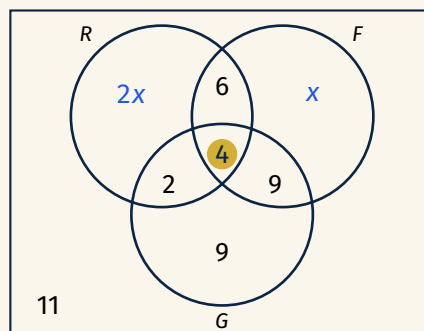
Fill the **centre** (all three) first, then each **pair-only** region (pair total minus the centre), then each **“only”** region. Some regions are in terms of x .

The regions **add up to the total**: that is the equation for x .

IT LOOKS LIKE

4 study all three

Russian only is twice French only



Jan 2021 P2H Q16, filled from the centre.

How do I do it?**1 Centre**

The number in all three.

2 Pair-only

Each “both A and B” total minus the centre.

3 Only regions

Set total minus its other three regions — or the x expressions.

4 Equation

All regions + outside = total. Solve, then answer the question.

ALL REGIONS = THE TOTAL That is your equation in x .

V6 Three Sets with x

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“10 study Russian and French”
(includes all three)

$$\rightarrow \text{Pair-only} = 10 - 4 = 6$$

“the number who study Russian only is
twice French only”

$$\rightarrow F \text{ only} = x, R \text{ only} = 2x$$

“Given that a total of 80 ...”

$$\rightarrow \text{Sum of all regions} = 80$$

WORKED EXAMPLE **PAST PAPER**

January 2021 · Paper 2H · Q16 · 6 marks

All three 4; R&F 10; F&G 13; R&G 6; G 24; none 11; Russian only is twice French only (x). 80 students. How many study Russian?

STEP 1 centre 4 ◀ all three

STEP 2 R&F only 6, F&G only 9, R&G only 2 ◀ pair totals - 4

STEP 3 G only = $24 - 4 - 9 - 2 = 9$, R only = $2x$ ◀ only regions

STEP 4 $2x + x + 6 + 4 + 2 + 9 + 9 + 11 = 80 \Rightarrow$ ◀ all regions = total
 $3x + 41 = 80 \Rightarrow x = 13$

ANSWER Russian = $2(13) + 6 + 4 + 2 = 38$

✓ regions add to 80 ✓

WATCH OUT

✗ putting 10 in the R&F region

✓ $10 - 4 = 6$

“Russian and French” counts the 4 who do all three too.

EXAM TIP

6 marks: diagram (3), equation (1), x (1), the final answer (1). Finish the question — here, Russian.

Check yourself

calculator first · answers on page 17

6.1 Jan 2020 P2HR

50 asked. D 28, C 18, R 20, C&R 8, D&R 9, D&C x , all three 6, none 5. Find x and $n(C' \cap D')$.

Answer:

6.2 Dr Eslam

60 people: all three 4; pair-only 6, 5, 3; A only x , B only $2x$, C only 10; none 8. Find x .

Answer:

6.3 Dr Eslam

Centre 2; each pair-only 3; A only x , B only x , C only $2x$; none 4; total 39. Find x .

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with Python set operations and sympy (check_answers.py).

1.1 {21, 27};
{21, 23, 24, 25, 27, 29}

1.2
{2, 3, 4, 6, 8, 9, 10, 12}

1.3
{1, 2, 4, 5, 7, 8, 10, 11}

2.1 No: 24 is not in E

2.2 No: e is in all three

2.3 $C = (A \cup B)'$

3.1 20

3.2 12

3.3 7

4.1 A only

4.2 outside both
circles

4.3 $A \cap B$ (the overlap)

5.1 7

5.2 both 7, only one 18

5.3 6

6.1 $x = 10$;
 $n(C' \cap D') = 14$

6.2 $x = 8$

6.3 $x = 6$

THE WHOLE TOPIC ON ONE PAGE

Sets & Venn Diagrams cheat sheet



Your notes
V1 List the members 8/25

$$A \cap B = \{9, 15\}$$

Write every set in full first.

V2 True? Explain 7/25

$$2, 3 \in A \cap B$$

Decision + a named member.

V3 Read a Venn 5/25

$$n(A \cup B) = 6 + 3 + 8 + 2$$

Shade in your head, then add.

V4 Shade a region

$$A \cap B' = A \text{ only}$$

 n keep overlap • ' swap.**V5 Fill a Venn 1/25**

$$22 - 7 = 15$$

Middle first, then only parts, then outside.

V6 Three sets 4/25

$$3x + 41 = 80$$

Centre $\rightarrow$ pairs $\rightarrow$ onlys; regions = total.**TOP 3 LOST MARKS**

1. a member not in $E \times$
2. a pair total put in the pair-only region $\times$
3. a reason with no member named $\times$

BOTTOM LINE n both • $\cup$ either • ' not • fill from the middle out • regions add to the total.